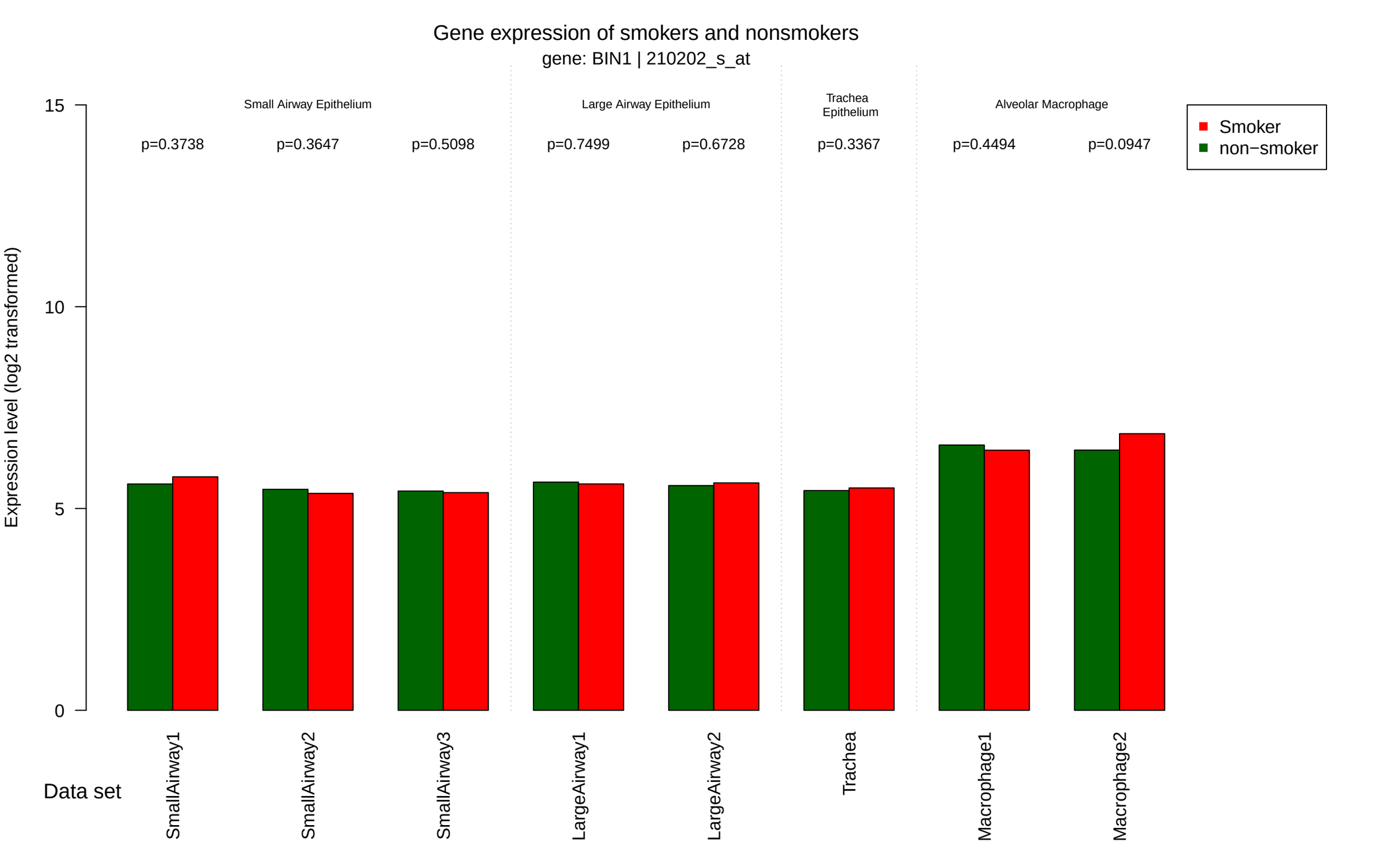
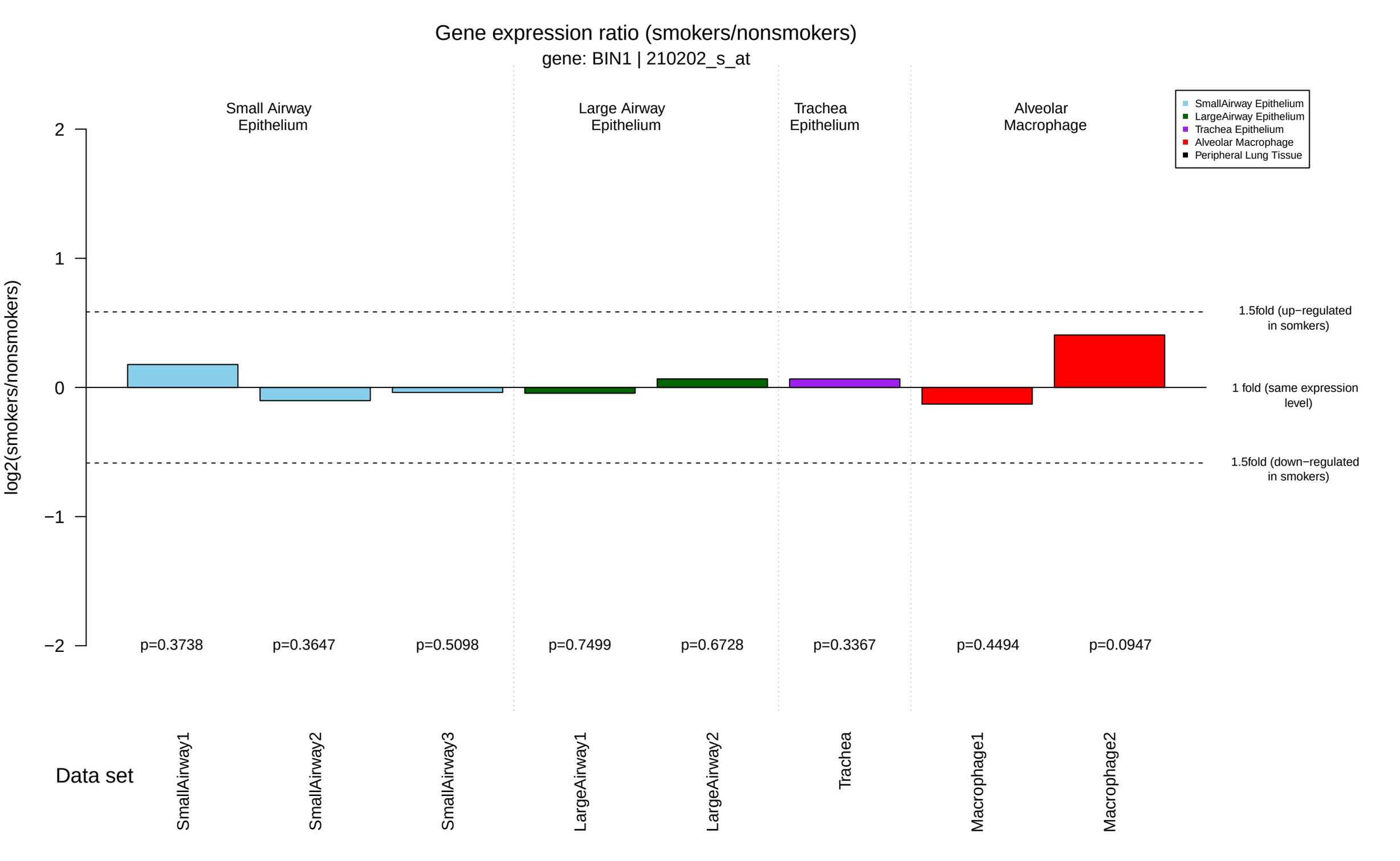
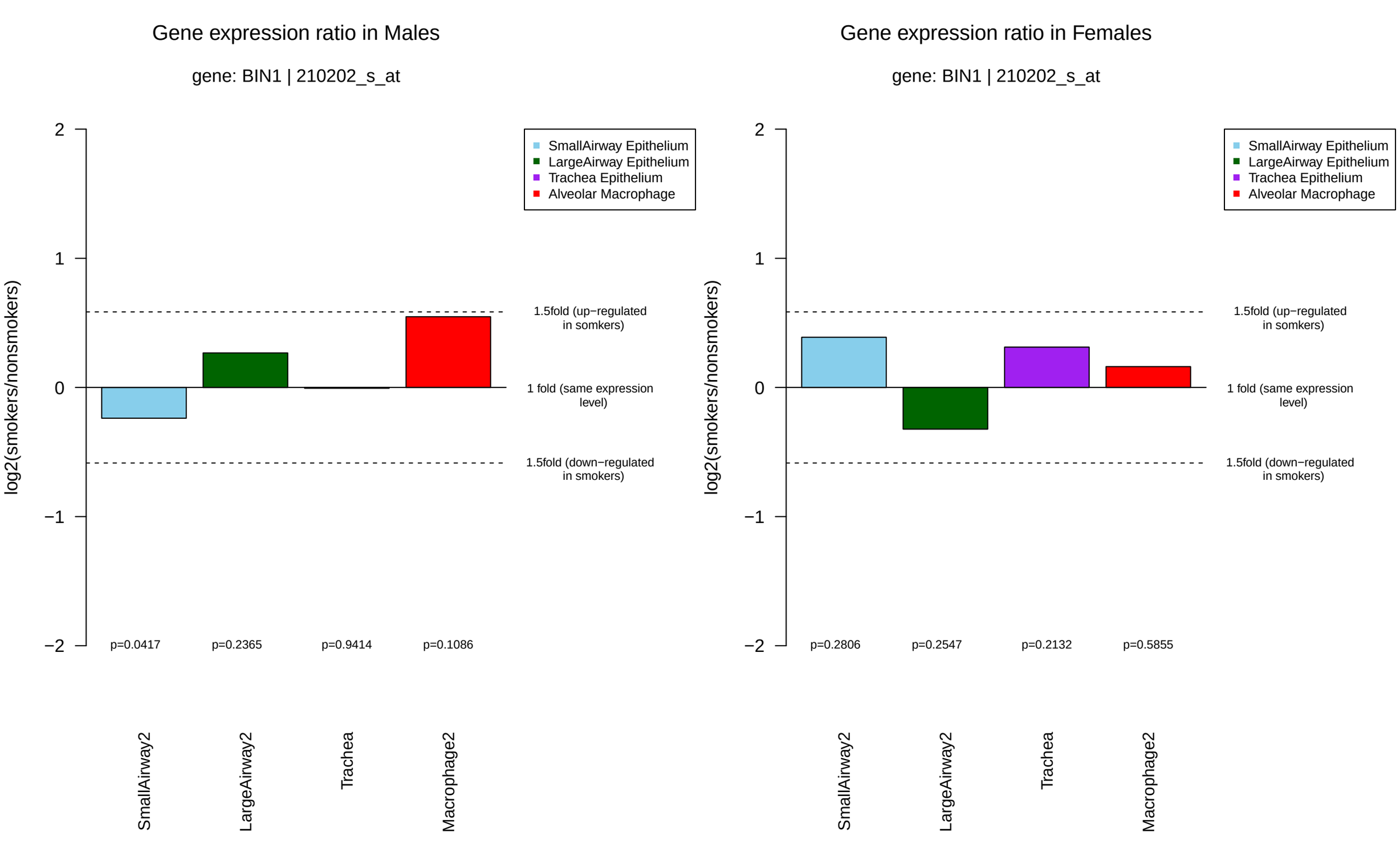
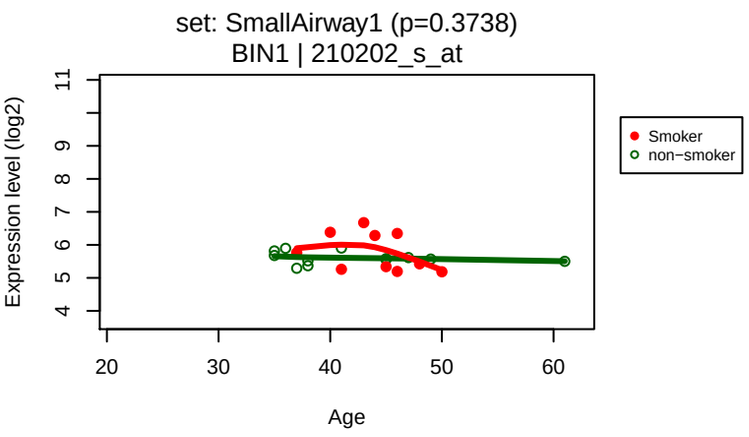
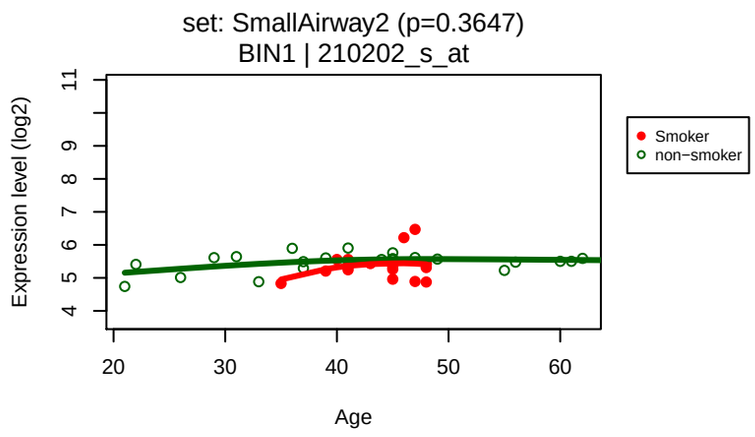
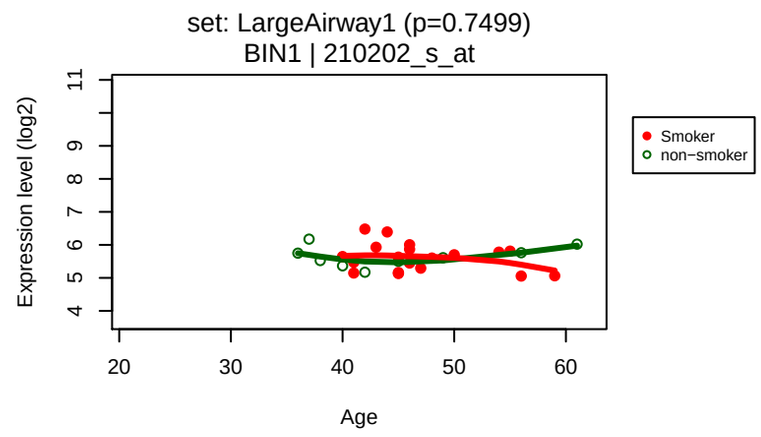

Gene expression of smokers and nonsmokers
gene: BIN1 | 210202_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3738
p=0.3647
p=0.5098
p=0.7499
p=0.6728
p=0.3367
p=0.4494
p=0.0947
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

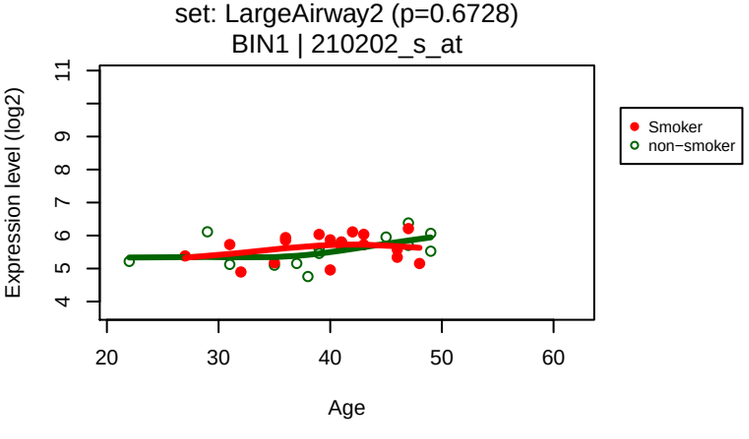

Gene expression ratio (smokers/nonsmokers)
gene: BIN1 | 210202_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3738
p=0.3647
p=0.5098
p=0.7499
p=0.6728
p=0.3367
p=0.4494
p=0.0947
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

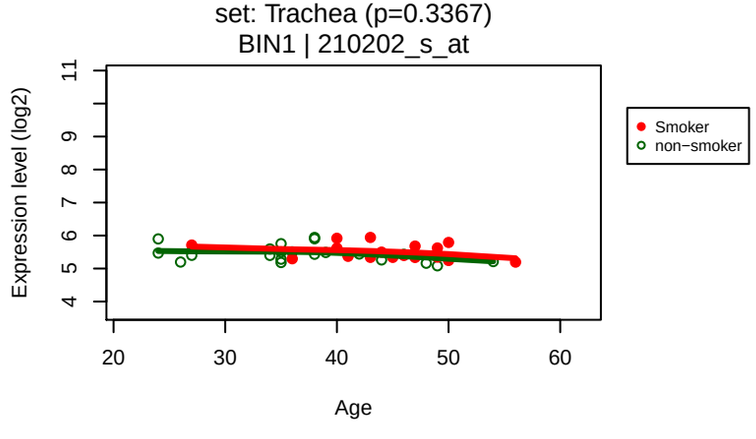

Gene expression ratio in Males
gene: BIN1 | 210202_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0417
p=0.2365
p=0.9414
p=0.1086
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BIN1 | 210202_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2806
p=0.2547
p=0.2132
p=0.5855
SmallAirway2
LargeAirway2
Trachea
Macrophage2

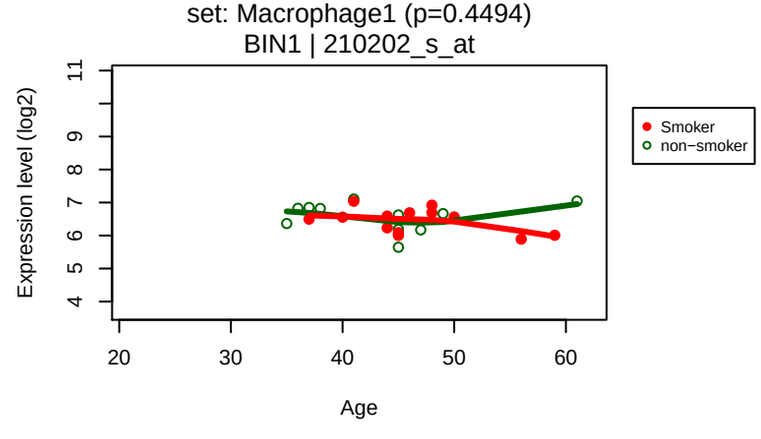

set: SmallAirway1 (p=0.3738)
BIN1 | 210202_s_at
Expression level (log2)
Age
Smoker
non−smoker

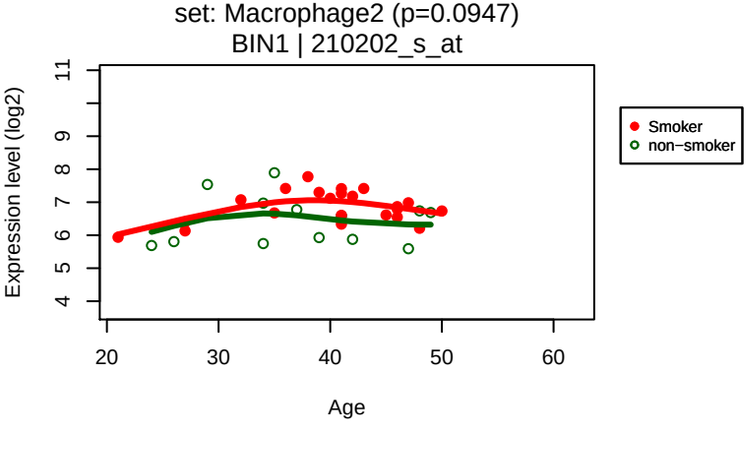

set: SmallAirway2 (p=0.3647)
BIN1 | 210202_s_at
Expression level (log2)
Age
Smoker
non−smoker

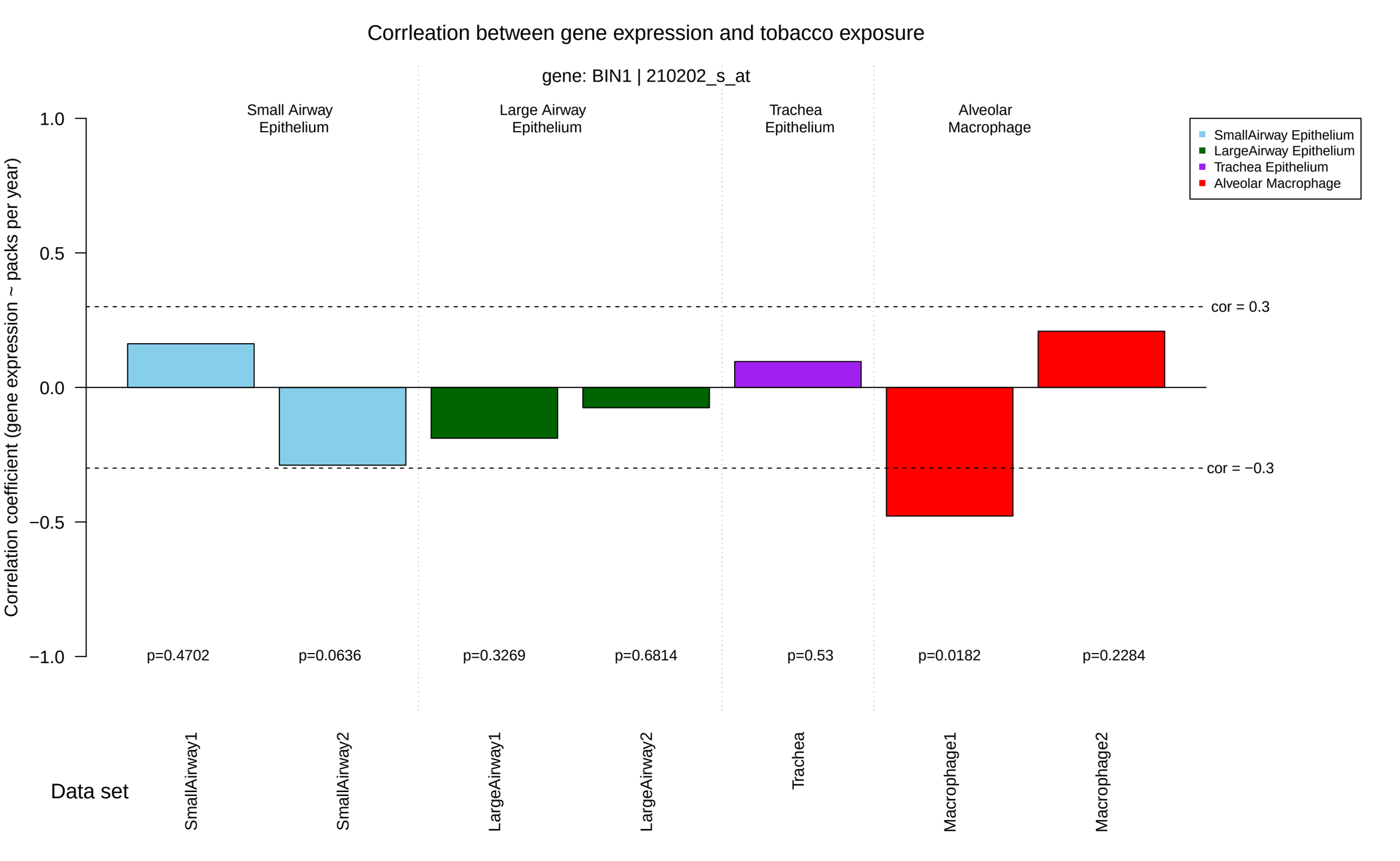

set: LargeAirway1 (p=0.7499)
BIN1 | 210202_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6728)
BIN1 | 210202_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3367)
BIN1 | 210202_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4494)
BIN1 | 210202_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0947)
BIN1 | 210202_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BIN1 | 210202_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4702
p=0.0636
p=0.3269
p=0.6814
p=0.53
p=0.0182
p=0.2284
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2